Relativistic Mean-Field Hadronic Models under Nuclear Matter Constraints

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Outline

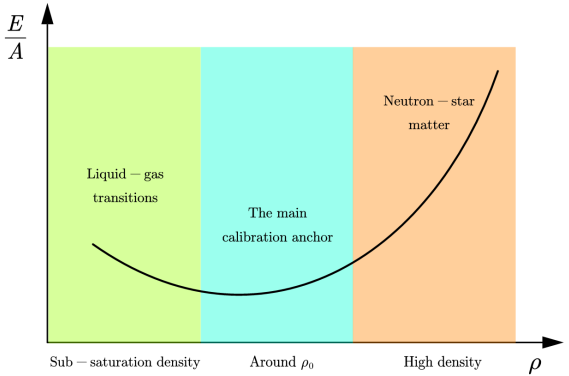
- Introduction
- Relativisitic Mean-field Models at Zero Temperature
- Results and Discussion
  - Previous constraints: SET1
  - Updated constraints: SET2a and SET2b
  - Excluding MIX3 constraint
  - Additional new constraints
- Summary and Conclusions

Introduction

No single quantitative theory covers both infinite nuclear matter and finite nuclei.

A sequence of effective descriptions

- **1935**: Bethe–Weizsäcker mass formula
- **1950s**: Nonrelativistic Skyrme models
- **1974**: Relativistic QHD / Walecka model



Agreement near ρ_0 does not guarantee agreement away from ρ_0 or in finite nuclei.

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Relativisitic Mean-field Models at Zero Temperature

More general nonlinear[2,3] Walecka[1] model:

$$\mathcal{L}_{\rm NL} = \mathcal{L}_{\rm nm} + \mathcal{L}_\sigma + \mathcal{L}_\omega + \mathcal{L}_\rho + \mathcal{L}_\delta + \mathcal{L}_{\sigma\omega\rho} \tag{1}$$

where

$$\mathcal{L}_{\rm nm} = \bar\psi\left(i\gamma^\mu\partial_\mu - M\right)\psi + g_\sigma\sigma\bar\psi\psi - g_\omega\bar\psi\gamma^\mu\omega_\mu\psi - \frac{g_\rho}{2}\bar\psi\gamma^\mu\vec\rho_\mu\vec\tau\psi + g_\delta\bar\psi\vec\delta\vec\tau\psi \tag{2a}$$

$$\mathcal{L}_\sigma = \frac{1}{2}\left(\partial^\mu\sigma\partial_\mu\sigma - m_\sigma^2\sigma^2\right) - \frac{A}{3}\sigma^3 - \frac{B}{4}\sigma^4 \tag{2b}$$

$$\mathcal{L}_\omega = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} + \frac{1}{2}m_\omega^2\omega_\mu\omega^\mu + \frac{C}{4}\left(g_\omega^2\omega_\mu\omega^\mu\right)^2 \tag{2c}$$

$$\mathcal{L}_\rho = -\frac{1}{4}\vec B^{\mu\nu}\vec B_{\mu\nu} + \frac{1}{2}m_\rho^2\vec\rho_\mu\vec\rho^\mu \tag{2d}$$

$$\mathcal{L}_\delta = \frac{1}{2}\left(\partial^\mu\vec\delta\partial_\mu\vec\delta - m_\delta^2\vec\delta^2\right) \tag{2e}$$

$$\mathcal{L}_{\sigma\omega\rho} = g_\sigma g_\omega^2\sigma\omega_\mu\omega^\mu\left(\alpha_1 + \frac{1}{2}\alpha_1' g_\sigma\sigma\right) + g_\sigma g_\rho^2\sigma\vec\rho_\mu\vec\rho^\mu\left(\alpha_2 + \frac{1}{2}\alpha_2' g_\sigma\sigma\right) + \frac{1}{2}\alpha_3' g_\omega^2 g_\rho^2\omega_\mu\omega^\mu\vec\rho_\mu\vec\rho^\mu \tag{2f}$$

and $F_{\mu\nu} = \partial_\nu\omega_\mu - \partial_\mu\omega_\nu$, $\vec B_{\mu\nu} = \partial_\nu\vec\rho_\mu - \partial_\mu\vec\rho_\nu - g_\rho\left(\vec\rho_\mu\times\vec\rho_\nu\right)$

Walecka model:
$\rho_0 \simeq -0.15 \sim 0.16\text{fm}^{-3}$
$E_0 \simeq -16\text{MeV}$
$M^\star/M \simeq 0.53 \sim 0.56\downarrow$
$K_\infty \simeq 546 \sim 625\text{MeV}\uparrow$

⇓

Nonliner Walecka model:
$K_\infty = 150 \pm 50\text{MeV}$
$M^\star/M \simeq 0.85$
EoS perform badly at high density.

⇓

Nonliner Walecka model:
ω^4 soften high-density EoS
$E_{\rm sym}$ and L perform badly.

⇓

δ meson and cross coupling...

[1] J. D. Walecka, Ann. Phys. 83, 491 (1974).
[2] J. Boguta and A. R. Bodmer, Nucl. Phys. A292, 413 (1977).
[3] Y. Sugahara, H. Toki, Nuclear Physics A579 (1994) 557-572

Relativisitic Mean-field Models at Zero Temperature

More general nonlinear[2,3] Walecka[1] model:

$$\mathcal{L}_{\rm NL} = \mathcal{L}_{\rm nm} + \mathcal{L}_\sigma + \mathcal{L}_\omega + \mathcal{L}_\rho + \mathcal{L}_\delta + \mathcal{L}_{\sigma\omega\rho} \tag{3}$$

where

$$\mathcal{L}_{\rm nm} = \bar\psi\left(i\gamma^\mu\partial_\mu - M\right)\psi + g_\sigma\sigma\bar\psi\psi - g_\omega\bar\psi\gamma^\mu\omega_\mu\psi - \frac{g_\rho}{2}\bar\psi\gamma^\mu\vec\rho_\mu\vec\tau\psi + g_\delta\bar\psi\vec\delta\vec\tau\psi \tag{4a}$$

$$\mathcal{L}_\sigma = \frac{1}{2}\left(\partial^\mu\sigma\partial_\mu\sigma - m_\sigma^2\sigma^2\right) - \frac{A}{3}\sigma^3 - \frac{B}{4}\sigma^4 \tag{4b}$$

$$\mathcal{L}_\omega = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} + \frac{1}{2}m_\omega^2\omega_\mu\omega^\mu + \frac{C}{4}\left(g_\omega^2\omega_\mu\omega^\mu\right)^2 \tag{4c}$$

$$\mathcal{L}_\rho = -\frac{1}{4}\vec B^{\mu\nu}\vec B_{\mu\nu} + \frac{1}{2}m_\rho^2\vec\rho_\mu\vec\rho^\mu \tag{4d}$$

$$\mathcal{L}_\delta = \frac{1}{2}\left(\partial^\mu\vec\delta\partial_\mu\vec\delta - m_\delta^2\vec\delta^2\right) \tag{4e}$$

$$\mathcal{L}_{\sigma\omega\rho} = g_\sigma g_\omega^2\sigma\omega_\mu\omega^\mu\left(\alpha_1 + \frac{1}{2}\alpha_1' g_\sigma\sigma\right) + g_\sigma g_\rho^2\sigma\vec\rho_\mu\vec\rho^\mu\left(\alpha_2 + \frac{1}{2}\alpha_2' g_\sigma\sigma\right) + \frac{1}{2}\alpha_3' g_\omega^2 g_\rho^2\omega_\mu\omega^\mu\vec\rho_\mu\vec\rho^\mu \tag{4f}$$

and $F_{\mu\nu} = \partial_\nu\omega_\mu - \partial_\mu\omega_\nu$, $\vec B_{\mu\nu} = \partial_\nu\vec\rho_\mu - \partial_\mu\vec\rho_\nu - g_\rho\left(\vec\rho_\mu\times\vec\rho_\nu\right)$

- **Type 1** (linear finite range models):
  $A = B = C = \alpha_1 = \alpha_2 = \alpha_1' = \alpha_2' = \alpha_3' = g_\delta = 0$
  H1, L1, L2, L3, RMF201-RMF206,...
- **Type 2** ($\sigma^3+\sigma^4$ models):
  $C = \alpha_1 = \alpha_2 = \alpha_1' = \alpha_2' = \alpha_3' = g_\delta = 0$
  NL3, GM1, FAMA1-FAMA5,...
- **Type 3** ($\sigma^3+\sigma^4+\omega_0^4$ models):
  $\alpha_1 = \alpha_2 = \alpha_1' = \alpha_2' = \alpha_3' = g_\delta = 0$
  PK1, TM1, TMA, NLSV1, NLSV2,...
- **Type 4** ($\sigma^3+\sigma^4+\omega_0^4$ + cross terms models):
  $g_\delta = 0$ and at least one of the coupling constants, $\alpha_1, \alpha_2, \alpha_1', \alpha_2', \alpha_3'$ is different from zero.
  BSR1-BSR21, Z27V1-Z271V6,...
- **Type 7** (Delta meson models):
  Parameterizations of finite range models presenting the meson δ, i.e., models in which $g_\delta \neq 0$.
  DD-MEδ, NLδ, NL3δ,...

[1] J. D. Walecka, Ann. Phys. 83, 491 (1974).
[2] J. Boguta and A. R. Bodmer, Nucl. Phys. A292, 413 (1977).
[3] Y. Sugahara, H. Toki, Nuclear Physics A579 (1994) 557-572

Relativisitic Mean-field Models at Zero Temperature

Consider mean-field approximation:

$$
\begin{aligned}
&\sigma \to \langle\sigma\rangle \equiv \sigma, && \omega_\mu \to \langle\omega_\mu\rangle \equiv \omega_0, \\
&\vec{\delta} \to \left\langle\vec{\delta}\right\rangle \equiv \delta_{(3)}, && \vec{\rho}_\mu \to \langle\vec{\rho}_\mu\rangle \equiv \bar{\rho}_{0(3)}, \\
&\rho_s = \langle\bar{\psi}\psi\rangle, && \rho_{s3} = \langle\bar{\psi}\tau_3\psi\rangle = \rho_{\mathrm{sp}} - \rho_{\mathrm{sn}}, \\
&\rho = \langle\bar{\psi}\gamma^0\psi\rangle = \rho_{\mathrm{p}} + \rho_{\mathrm{n}}, && \rho_3 = \langle\bar{\psi}\gamma^0\tau_3\psi\rangle = \rho_{\mathrm{p}} - \rho_{\mathrm{n}} = (2y-1)\,\rho
\end{aligned}
\tag{5}
$$

The energy density and the pressure of the asymmetric system:

$$
\begin{aligned}
\mathcal{E}_{\mathrm{NL}} = &\sum_\tau \left(\frac{3}{4}E_\tau\rho_\tau + \frac{1}{4}M^\star_\tau\rho_{s\tau}\right) + \frac{1}{2}m_\sigma^2\sigma^2 + \frac{1}{3}A\sigma^3 + \frac{1}{4}B\sigma^4 - \frac{1}{2}m_\omega^2\omega_0^2 - \frac{1}{4}C\left(g_\omega^2\omega_0^2\right)^2 \\
&- \frac{1}{2}m_\rho^2\bar{\rho}_{0(3)}^2 + g_\omega\omega_0\rho + \frac{1}{2}g_\rho\bar{\rho}_{0(3)}\rho_3 + \frac{1}{2}m_\delta^2\delta_{(3)}^2 - g_\sigma g_\omega^2\sigma\omega_0^2\left(\alpha_1 + \frac{1}{2}\alpha_1' g_\sigma\sigma\right) \\
&- g_\sigma g_\rho^2\sigma\bar{\rho}_{0(3)}^2\left(\alpha_2 + \frac{1}{2}\alpha_2' g_\sigma\sigma\right) - \frac{1}{2}\alpha_3' g_\omega^2 g_\rho^2\omega_0^2\bar{\rho}_{0(3)}^2
\end{aligned}
\tag{6}
$$

$$
\begin{aligned}
P_{\mathrm{NL}} = &\sum_\tau \left(\frac{1}{4}E_{F\tau}\rho_\tau - \frac{1}{4}M^\star_\tau\rho_{s\tau}\right) - \frac{1}{2}m_\sigma^2\sigma^2 - \frac{1}{3}A\sigma^3 - \frac{1}{4}B\sigma^4 + \frac{1}{2}m_\omega^2\omega_0^2 + \frac{1}{4}C\left(g_\omega^2\omega_0^2\right)^2 \\
&+ \frac{1}{2}m_\rho^2\bar{\rho}_{0(3)}^2 + \frac{1}{2}\alpha_3' g_\omega^2 g_\rho^2\omega_0^2\bar{\rho}_{0(3)}^2 - \frac{1}{2}m_\delta^2\delta_{(3)}^2 + g_\sigma g_\omega^2\sigma\omega_0^2\left(\alpha_1 + \frac{1}{2}\alpha_1' g_\sigma\sigma\right) \\
&+ g_\sigma g_\rho^2\sigma\bar{\rho}_{0(3)}^2\left(\alpha_2 + \frac{1}{2}\alpha_2' g_\sigma\sigma\right)
\end{aligned}
\tag{7}
$$

Relativisitic Mean-field Models at Zero Temperature

Another widely used approach: density dependence in the aforementioned couplings. Its Lagrangian density:

$$\begin{aligned}\mathcal{L}_{\mathrm{DD}} =& \bar{\psi}\left(i\gamma^{\mu}\partial_{\mu} - M\right)\psi + \Gamma_{\sigma}\left(\rho\right)\sigma\bar{\psi}\psi - \Gamma_{\omega}\left(\rho\right)\bar{\psi}\gamma^{\mu}\omega_{\mu}\psi - \frac{1}{2}\Gamma_{\rho}\left(\rho\right)\bar{\psi}\gamma^{\mu}\vec{\rho}_{\mu}\vec{\tau}\psi + \Gamma_{\delta}\left(\rho\right)\bar{\psi}\vec{\delta}\vec{\tau}\psi \\ &+ \frac{1}{2}\left(\partial^{\mu}\sigma\partial_{\mu}\sigma - m_{\sigma}^{2}\sigma^{2}\right) - \frac{1}{4}F^{\mu\nu}F_{\mu\nu} + \frac{1}{2}m_{\omega}^{2}\omega_{\mu}\omega^{\mu} - \frac{1}{4}\vec{B}^{\mu\nu}\vec{B}_{\mu\nu} + \frac{1}{2}m_{\rho}^{2}\vec{\rho}_{\mu}\vec{\rho}^{\mu} \\ &+ \frac{1}{2}\left(\partial^{\mu}\vec{\delta}\partial_{\mu}\vec{\delta} - m_{\delta}^{2}\vec{\delta}^{2}\right) \end{aligned} \tag{8}$$

where

$$\Gamma_{\sigma,\omega}\left(\rho\right) = \Gamma_{\sigma,\omega}\left(\rho_0\right) f_{\sigma,\omega}\left(x\right), \quad \Gamma_{\rho}\left(\rho\right) = \Gamma_{\rho}\left(\rho_0\right) e^{-a(x-1)}, \quad f_{\sigma,\omega}\left(x\right) = a_{\sigma,\omega}\frac{1 + b_{\sigma,\omega}\left(x + d_{\sigma,\omega}\right)^{2}}{1 + c_{\sigma,\omega}\left(x + d_{\sigma,\omega}\right)^{2}}, \quad x = \frac{\rho}{\rho_0} \tag{9}$$

The energy density and pressure are given by:

$$\mathcal{E}_{\mathrm{DD}} = \sum_{\tau}\left(\frac{3}{4}E_{\tau}\rho_{\tau} + \frac{1}{4}M_{\tau}^{\star}\rho_{s\tau}\right) + \frac{1}{2}m_{\sigma}^{2}\sigma^{2} - \frac{1}{2}m_{\omega}^{2}\omega_{0}^{2} - \frac{1}{2}m_{\rho}^{2}\bar{\rho}_{0(3)}^{2} + \frac{1}{2}m_{\delta}^{2}\delta_{(3)}^{2} + \Gamma_{\omega}\left(\rho\right)\omega_{0}\rho + \frac{1}{2}\Gamma_{\rho}\left(\rho\right)\bar{\rho}_{0(3)}\rho_{3} \tag{10}$$

$$\begin{aligned}P_{\mathrm{DD}} =& \sum_{\tau}\left(\frac{1}{4}E_{F\tau}\rho_{\tau} - \frac{1}{4}M_{\tau}^{\star}\rho_{s\tau}\right) + \rho\left(\frac{\partial\Gamma_{\omega}}{\partial\rho}\omega_{0}\rho + \frac{1}{2}\frac{\partial\Gamma_{\rho}}{\partial\rho}\bar{\rho}_{0(3)}\rho_{3} - \frac{\partial\Gamma_{\sigma}}{\partial\rho}\sigma\rho_{s} - \frac{\partial\Gamma_{\delta}}{\partial\rho}\delta_{(3)}\rho_{s3}\right) - \frac{1}{2}m_{\sigma}^{2}\sigma^{2} \\ &+ \frac{1}{2}m_{\omega}^{2}\omega_{0}^{2} + \frac{1}{2}m_{\rho}^{2}\bar{\rho}_{0(3)}^{2} - \frac{1}{2}m_{\delta}^{2}\delta_{(3)}^{2}\end{aligned} \tag{11}$$

- **Type 5** (Density Dependent models[4]):
  Parameterizations obtained from Eq. (8) in which $\Gamma_{\delta} = 0$.
  DD-F, DD-ME1, DD-M32, PKDD,...

[4] Typel, S. and Wolter, H.H., Nucl. Phys. A656 ,331 (1999)

The nonlinear point-coupling (NLPC) model[5]:

$$\begin{aligned}\mathcal{L}_{\mathrm{NLPC}} =& \bar{\psi}\left(i\gamma^{\mu}\partial_{\mu} - M\right)\psi - \frac{1}{2}\alpha_s\left(\bar{\psi}\psi\right)^2 - \frac{1}{3}\beta_s\left(\bar{\psi}\psi\right)^3 - \frac{1}{4}\gamma_s\left(\bar{\psi}\psi\right)^4 - \frac{1}{2}\alpha_V\left(\bar{\psi}\gamma^{\mu}\psi\right)^2 - \frac{1}{4}\gamma_V\left(\bar{\psi}\gamma^{\mu}\psi\right)^4 \\ & - \frac{1}{2}\alpha_{TV}\left(\bar{\psi}\gamma^{\mu}\vec{\tau}\psi\right)^2 - \frac{1}{4}\gamma_{TV}\left(\bar{\psi}\gamma^{\mu}\vec{\tau}\psi\right)^4 - \frac{1}{2}\alpha_{TS}\left(\bar{\psi}\vec{\tau}\psi\right)^2 + \left[\eta_1 + \eta_2\left(\bar{\psi}\psi\right)\right]\left(\bar{\psi}\psi\right)\left(\bar{\psi}\gamma^{\mu}\psi\right)^2 \\ & - \eta_3\left(\bar{\psi}\psi\right)\left(\bar{\psi}\gamma^{\mu}\vec{\tau}\psi\right)^2 \end{aligned} \tag{12}$$

The energy density and pressure are given by:

$$\begin{aligned}\mathcal{E}_{\mathrm{NLPC}} =& \sum_{\tau}\left(\frac{3}{4}E_{\tau}\rho_{\tau} + \frac{1}{4}M^{\star}_{\tau}\rho_{s\tau}\right) + \frac{1}{2}\alpha_V\rho^2 + \frac{1}{2}\alpha_{TV}\rho_3^2 + \frac{1}{4}\gamma_V\rho^4 + \frac{1}{4}\gamma_{TV}\rho_3^4 - \eta_2\rho_s^2\rho^2 \\ & - \frac{1}{2}\alpha_s\rho_s^2 - \frac{2}{3}\beta_s\rho_s^3 - \frac{3}{4}\gamma_s\rho_s^4 - \frac{1}{2}\alpha_{TS}\rho_{s3}^2 \end{aligned} \tag{13}$$

$$\begin{aligned}P_{\mathrm{NLPC}} =& \sum_{\tau}\left(\frac{1}{4}E_{F\tau}\rho_{\tau} - \frac{1}{4}M^{\star}_{\tau}\rho_{s\tau}\right) + \frac{1}{2}\alpha_V\rho^2 + \frac{1}{2}\alpha_{TV}\rho_3^2 + \frac{3}{4}\gamma_V\rho^4 + \frac{3}{4}\gamma_{TV}\rho_3^4 + 2\eta_1\rho_s\rho^2 \\ & + 3\eta_2\rho_s^2\rho^2 + 2\eta_3\rho_s\rho_3^2 + \frac{1}{2}\alpha_s\rho_s^2 + \frac{2}{3}\beta_s\rho_s^3 + \frac{3}{4}\gamma_s\rho_s^4 + \frac{1}{2}\alpha_{TS}\rho_{s3}^2 \end{aligned} \tag{14}$$

- **Type 6** (Point-coupling models):
  parameterizations of the model described by Eq. (12) in which $\alpha_{TS} = 0$.
  PC-F1, PC-F3,...

[5] B. A. Nikolaus, T. Hoch, and D. G. Madland, Phys. Rev. C 46, 1757 (1992)

Relativisitic Mean-field Models at Zero Temperature

Several quantities chosen as constraints that are presented in the next section are directly calculated from $\mathcal{E}, P$:

- The symmetry energy (general definition): $\mathcal{S} = \frac{1}{8} \left.\frac{\partial^2(\mathcal{E}/\rho)}{\partial y^2}\right|_{y=\frac{1}{2}}$
- Incompressibility: $K_0 = 9 \left.\frac{\partial P}{\partial \rho}\right|_{\rho=\rho_0, y=\frac{1}{2}}$
- Skewness coefficient: $Q_0 = 27\rho_0^3 \left.\frac{\partial^3(\mathcal{E}/\rho)}{\partial \rho^3}\right|_{\rho=\rho_0, y=\frac{1}{2}}$
- The symmetry energy:$J = \mathcal{S}(\rho_0)$
- Slope of symmetry energy: $L_0 = 3\rho_0 \left.\frac{\partial \mathcal{S}}{\partial \rho}\right|_{\rho=\rho_0}$
- Curvature of symmetry $K_{\rm sym}^0 = 3\rho_0^2 \left.\frac{\partial^2 \mathcal{S}}{\partial \rho^2}\right|_{\rho=\rho_0}$
- Skewness of $\mathcal{S}$: $Q_{\rm sym}^0 = 27\rho_0^3 \left.\frac{\partial^3 \mathcal{S}}{\partial \rho^3}\right|_{\rho=\rho_0}$
- Volume part of the isospin incompressibility: $K_{\tau v}^0 = \left(K_{\rm sym}^0 - 6L_0 - \frac{Q_0}{K_0}L_0\right)$

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Previous constraints: SET1

Type 1-7 form **263** relativistic models, which will be analysed with two sets of constraints.
The first set of constraints (SET1) are shown in table below.

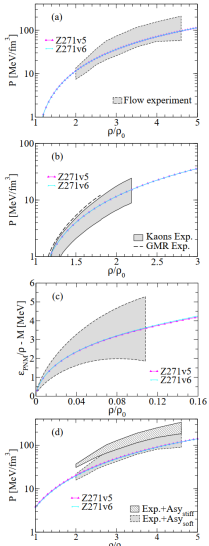
A model is considered approved in a numerical constraint if

$$|\mathrm{Dev}| = \left| \frac{Q_{\mathrm{mod}} - Q_{\mathrm{const}}}{\Delta} \right| \leq 1 \qquad (15)$$

and the model is inside the corresponding band in 95% or more of the density region.

Constraint	Quantity	Density Region	Range of constraint (exp/emp)	Range of constraint from CRMF
SM1	K_0	ρ_0 (fm^{-3})	200 – 260 MeV	271.0 MeV
SM2	$K' = -Q_0$	ρ_0 (fm^{-3})	200 – 1200 MeV	733.6 MeV
SM3	$P(\rho)$	$2 < \rho/\rho_0 < 4.6$	Band Region	right figure
SM4	$P(\rho)$	$1.2 < \rho/\rho_0 < 2.2$	Band Region	right figure
PNM1	$\mathcal{E}_{\mathrm{PNM}}/\rho$	$0.017 < \rho/\rho_0 < 0.108$	Band Region	right figure
PNM2	$P(\rho)$	$2 < \rho/\rho_0 < 4.6$	Band Region	right figure
MIX1	J	ρ_0 (fm^{-3})	30 – 35 MeV	33.8 – 34.0 MeV
MIX2	L_0	ρ_0 (fm^{-3})	40 – 76 MeV	70.9 – 73.9 MeV
MIX3	$K^0_{\tau,\mathrm{v}}$	ρ_0 (fm^{-3})	−760 – −372 MeV	−388.5 – −388.4 MeV
MIX4	$\mathcal{S}(\rho_0/2)/J$	ρ_0 (fm^{-3})	0.57 – 0.86	0.58
MIX5	$3P_{\mathrm{PNM}}/L_0\rho_0$	ρ_0 (fm^{-3})	0.90 – 1.10	1.05 – 1.06

Table: SET1 constraints used in the CRMF fit.



None of the models satisfies all constraints simultaneously!

Previous constraints: SET1

There are 12 parameterzations that do not satisfy only one constraint. Z271v5 and Z271v6, fall outside the limits of the SM1 constraint by less than 5%.

Model	Model value (MeV)	Deviation
constraint not satisfied: SM1 ($200 \leqslant K_0 \leqslant 260$ MeV)		
Z271v5	271.00	1.37
Z271v6	271.00	1.37
constraint not satisfied: MIX3 ($-760 \leqslant K^0_{\tau,\mathrm{v}} \leqslant -372$ MeV)		
BSR15	-252.54	1.62
BSR16	-258.75	1.58
FSUGold	-276.07	1.49
FSUGZ06	-259.47	1.58
FSUGold4	-205.59	1.86
FSU-III	-341.03	1.16
FSU-IV	-210.68	1.83
TW99	-332.32	1.20
DD-F	-285.54	1.45
DD-ME6	-258.28	1.59

Table: Models violating the selected constraints.

Constraints	SM1	SM2	SM3	SM4	PNM1	PNM2
Number of models	146	174	104	153	193	101
Constraints	MIX1	MIX2	MIX3	MIX4	MIX5	
Number of models	162	59	124	65	258	

Table: Number of models satisfying each individual constraint.

MIX5 constraint is the "weakest" one,
while MIX2 is the "strongest" one.

Define Z271v5 and Z271v6 as included in the Consistent Relativistic Mean-Field (CRMF) parameterizations, i.e., the ones satisfying the constraints of SET1.

Updated constraints: SET2a and SET2b

Why update SET1?

- SET1 is used to analyse the Skyrme-type parameterizations[6]. The theoretical framework of RMF is different from Skyrme.
  Several studies involving both RMF models and non-relativistic models have shown that although both sets of models verify the same correlations, the parameter distribution of each set is not completely overlapping.
- The constraints imposed within each of the sets are not measured directly, but they result from the analysis of raw data, which involves model assumptions.
  e.g. $E_{\mathrm{GMR}}^{\mathrm{expt}} \xrightarrow{\text{correlation}} K_0 \Rightarrow K_0 \simeq 230\mathrm{MeV}$, K_0 constraint is model dependent and contains several errors.

Ways to update SET1:

- Based on new insights into the relationship between K_0 and GMR, $K_0 = 230 \pm 30$ MeV $\Rightarrow 230 \pm 40$ MeV.
  Reconsider J, L and $K_{\tau,v}$.
- Remove SM2 ($200 \leq K' \leq 1200$ MeV).
  K' constraint is entirely based on parameterizations of the nonrelativistic Skyrme model.
- Recent study generates a new constraint that EoS must satisfy for the mass-radius relation of neutron stars.
  $\Rightarrow$ Modify SM3 constraint, i.e. the performance of $P(\rho)$ in high-density region of SNM.
  SM3a: Old SM3 constraint $\rightarrow$ SET2a
  SM3b: New SM3 constraint $\rightarrow$ SET2b
- Remove PNM2 constraint, i.e. the performance of $P(\rho)$ in high-density region of PNM.
  Excessive model dependence and no new experimental information on heavy-ion collisions.
- Remove MIX5. Too weak to constraint.

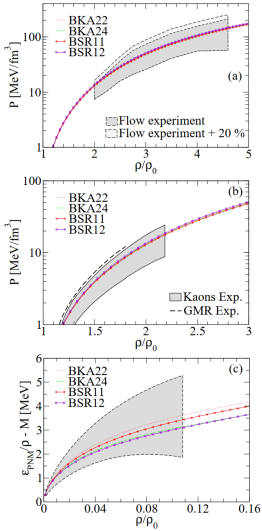
[6] M. Dutra, O. Louren,co, et al, Phys. Rev. C 85, 035201 (2012).

Updated constraints: SET2a and SET2b

SET2a and SET2b are shown in table below:

Constraint	Quantity	Density Region	Range of constraint (exp/emp)	Range of constraint from CRMF
SM1	K_0	ρ_0 (fm^{-3})	$190 - 270$ MeV	$225.2 - 232.4$ MeV
SM3a	the same as SM3b plus 20% on upper limit			right figure
SM3b	$P(\rho)$	$2 < \frac{\rho}{\rho_0} < 5$	Band Region	right figure
SM4	$P(\rho)$	$1.2 < \frac{\rho}{\rho_0} < 2.2$	Band Region	right figure3
PNM1	$\varepsilon_{\rm PNM}/\rho$	$0.017 < \frac{\rho}{\rho_0} < 0.108$	Band Region	right figure
MIX1a	J	ρ_0 (fm^{-3})	$25 - 35$ MeV	$33.2 - 34.2$ MeV
MIX1b	J	ρ_0 (fm^{-3})	$30 - 35$ MeV	$33.2 - 34.0$ MeV
MIX2a	L_0	ρ_0 (fm^{-3})	$25 - 115$ MeV	$77.9 - 84.8$ MeV
MIX2b	L_0	ρ_0 (fm^{-3})	$30 - 80$ MeV	$77.9 - 78.8$ MeV
MIX3	$K^0_{\tau,\nu}$	ρ_0 (fm^{-3})	$-700 - -400$ MeV	-421.6(a)$/-414.3$(b)$--382.5$ M
MIX4	$\frac{\mathcal{S}(\rho_0/2)}{J}$	ρ_0 (fm^{-3})	$0.57 - 0.86$	0.57(a)$/0.59$(b)$- 0.59$

Table: SET1 constraints used in the CRMF fit.



Update constraints: SET2a and SET2b

As before, apply SET2a and SET2b to 263 relativistic models

Model	approved in SET	ρ_0 (fm^{-3})	E_0 (MeV)	K_0 (MeV)	m^*	K' (MeV)	J (MeV)	L_0 (MeV)	$K^0_{\tau,\mathrm{v}}$ (MeV)	$\frac{S(\rho_0/2)}{J}$
BKA22	2a and 2b	0.147	−15.91	225.24	0.61	283.29	33.17	78.79	**-382.46**	0.59
BKA24	2a	0.147	−15.95	227.06	0.60	273.58	34.19	**84.80**	−421.55	0.57
BSR11	2a and 2b	0.147	−16.08	226.75	0.61	312.37	33.69	78.78	**-388.86**	0.59
BSR12	2a and 2b	0.147	−16.10	232.35	0.61	290.31	34.00	77.90	−414.30	0.59
Z271v5	1	0.148	−16.24	**271.00**	0.80	733.59	34.04	73.90	−388.52	0.58
Z271v6	1	0.148	−16.24	**271.00**	0.80	733.59	33.80	70.94	−388.36	0.58

Table: Nuclear matter properties at the saturation density ρ_0 of the RMF models consistent with the macroscopic constraints.

The number of approved models in each constraint:

	SET2										
Constraints	SM1	SM3a	SM3b	SM4	PNM1	MIX1a	MIX1b	MIX2a	MIX2b	MIX3	MIX4
Number of models	153	129	104	153	193	174	162	216	72	96	65

Table: Number of approved models (among 263) in each constraint of SET2.

BKA22, BKA24, BSR11, BSR12 belong to Type 4 ($\sigma^3 + \sigma^4 + \omega_0^4 +$ cross terms models)

Excluding MIX3 constraint

If we simply discard the MIX3 constraint from SET2a and SET2b,

- The new results for the **SET2a** analysis become the following,
  - **25** models are consistent with all 7 constraints:
    BKA20, BKA22, BKA24, BSR8, BSR9, BSR10, BSR11, BSR12, BSR15, BSR16, BSR17, BSR18, BSR19, DD-MEδ, DD-F, DDHδ, FSU-III, FSU-IV, FSUGold, FSUGold4, FSUGZ03, FSUGZ06, G2*, IU-FSU, and TW99.
  - 48 models satisfy all but one of the constraints. In this group, **10** models fell outside the range of the constraint by less than 5%. They are: BSR20, FA3, Z271s2, Z271s3, Z271s4, Z271s5, Z271s6, Z271v4, Z271v5, and Z271v6. By also including such models in the CRMF parameterizations, one has **35** models consistent with the SET2a analysis.
- The new results for the **SET2b** analysis become the following,
  - **22** models consistent with all constraints. These are the same models approved in SET2a, except for the BKA24, IU-FSU and DDHδ models.
  - 14 models satisfy only 7 constraints, and by applying the 5% criterium, **8** more models are approved. They are the same as in the corresponding case of SET2a, except for the BSR20 and FA3 models. Therefore, one has in the SET2b analysis a total of **30** models in the group of approved models.

Additional new constraints

New constraints from reanalysis of data on GMR energies became available

$$\text{SM1 } (K_0)\text{: 190-270MeV} \Rightarrow \text{SM1-new: 250-315 MeV}$$
$$\text{MIX3 } (K_{\tau,\nu})\text{:-700–400MeV} \Rightarrow \text{MIX3-new: -620–370MeV}$$

Re-analysed the data introducing the two additional constraints with the following results:

- Analysis considering MIX3-new in SET2
  - **SET2a** (SM1-new, SM3a, SM4, PNM1, MIX1a, MIX2a, MIX3-new and MIX4): **4** models approved, namely, Z271v4, Z271v5, Z271v6, and FA3. 24 models not satisfying only one constraint. Even with 5% tolerance, no further model approved.
  - **SET2b** (SM1-new, SM3b, SM4, PNM1, MIX1b, MIX2b, MIX3-new and MIX4): **3** models approved, namely, Z271v4, Z271v5, and Z271v6. 12 models not satisfying only one constraint. Even with 5% tolerance, no further model approved.
- Analysis discarding MIX3-new in SET2
  - **SET2a** (SM1-new, SM3a, SM4, PNM1, MIX1a, MIX2a, and MIX4): **9** models approved, namely, Z271v4, Z271v5, Z271v6, Z271s2, Z271s3, Z271s4, Z271s5, Z271s6, and FA3. 45 models not satisfying only one constraint. Even with 5% tolerance, no further model approved.
  - **SET2b** (SM1-new, SM3b, SM4, PNM1, MIX1b, MIX2b, and MIX4): **8** models approved, namely, Z271v4, Z271v5, Z271v6, Z271s2, Z271s3, Z271s4, Z271s5, and Z271s6. 26 models not satisfying only one constraint. Even with 5% tolerance, no further model approved.

Outline

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Summary and Conclusions

- **Systematic assessment of RMF models**: A total of 263 RMF parameterizations were tested against constraints from symmetric nuclear matter, pure neutron matter, and symmetry-energy properties.
- **SET1 is highly restrictive**: No parameterization strictly satisfies all SET1 constraints. With a 5% tolerance, only Z271v5 and Z271v6 are classified as consistent RMF models.
- **Updated constraints lead to SET2**: By incorporating updated experimental, theoretical, and astrophysical information and reconsidering model dependence, the authors construct SET2a and SET2b, yielding 4 and 3 consistent RMF parameterizations, respectively.
- $K_{\tau,v}$ **is a critical constraint**: The MIX3 constraint strongly affects the selection. Once it is removed, the number of accepted models increases from 4 and 3 to 35 and 30, indicating significant uncertainty and model dependence in $K_{\tau,v}$.
- **Cross couplings are important**: All models satisfying the complete SET2 constraints belong to Type 4, which contains σ-ω-ρ cross couplings. This highlights the importance of a flexible isovector density dependence for describing the symmetry energy.
- **Nuclear-matter constraints are not sufficient**: Agreement with infinite nuclear matter does not guarantee a good description of finite nuclei. A unified protocol combining nuclear-matter properties and finite-nucleus observables, such as binding energies, charge and neutron radii, isotope shifts, and spin-orbit splittings, is therefore desirable.
- The selected consistent RMF models are preferable for further nuclear and neutron-star applications, but improved experimental constraints and more unified fitting strategies are still required.

Thanks for your attention!